COLD FORMED HOLLOW SECTIONS

Standard Specification	Scope	Grade of Tubes	Chemical Composition Percentage (max. unless range is given)				Mechanical Properties				
			C	Mn	P	S	Condition (min.)	Tensile Strength (min.)	Yield Strength (% min.)	Elongation	
1	2	3	4	5	6	7	8	9	10	11	
								kgf/mm²	kgf/mm²	longitudinal	transversal
										No. 11,12 T.P.	No. 5 T.P.
JIS-G 3444	E.R.W. Carbon Steel Pipes for Civil										
	Engineer Architecture,	STK - 30	--	--	0.05	0.05	As Rolled	30	_	30	25
	Steel Towers, Scaffol-	STK - 41	0.25	--	0.04	0.04		41	24	23	18
	ding, Pile Struts &	STK - 51	0.30	0.3 - 1.0	0.04	0.04		51	36	15	10
	other Structures.	STK - 50	0.18	1.50	0.04	0.04		50	32	23	18
										No. 4,11,12 T.P.	No. 4,5 T.P.
JIS-G 3445	E.R.W. Carbon Steel	STKM-11A	0.12	0.25-0.6	0.04	0.04	As Rolled	30	--	35	30
	Pipes for Machine	STKM-12A	0.20	0.25-0.6	0.04	0.04		35	18	35	30
	Structural Purposes	STKM-12B						40	28	25	20
		STKM-12C						48	36	20	15
		STKM-13A	0.25	0.30-0.9	0.04	0.04		38	22	30	25
		STKM-13B						45	31	20	15
		STKM-13C	0.30	0.30-1.0	0.04	0.04		52	39	15	10
		STKM-14A						42	25	25	20
		STKM-14B						51	36	15	10
										No. 11, 12 T.P	No. 5 T.P.
JIS-G 3452	E.R.W. Carbon Steel Pipes for ordinary use	SGP	--	--	0.05	0.05	As Rolled (Black Tubes)	30	--	30	25
										No. 5 T.P	--
JIS-G 3466	E.R.W. Carbon Steel	STKR-41	0.25	--	0.04	0.04	As Rolled	41	25	23	--
	Square Tubes for	STKR-50	0.18	1.50	0.04	0.04		50	33	23	--
	General Structural Purposes										

*1: G.L. = Gauge Length
*2: S_o = Original Cross-Section Area
*3: T.S. = Tensile Strength
*4: T.P. = Test Piece as per JIS Z2201
*5: H = Distance between Outside Surfaces
*6: D = Outside Diameter
*7: t = Wall Thickness of Tubes
*8: H′ = Distance between Inside Surfaces